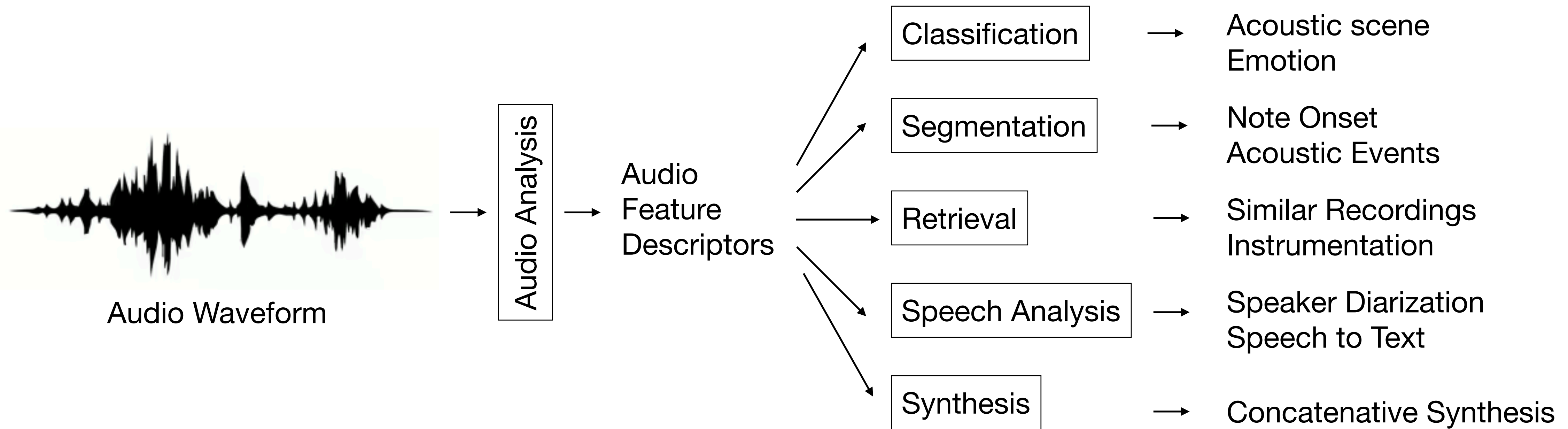


Audio Feature Descriptors

Audio Feature Descriptors - Pipeline



Audio Feature Descriptors - Representation Domains

Time Domain

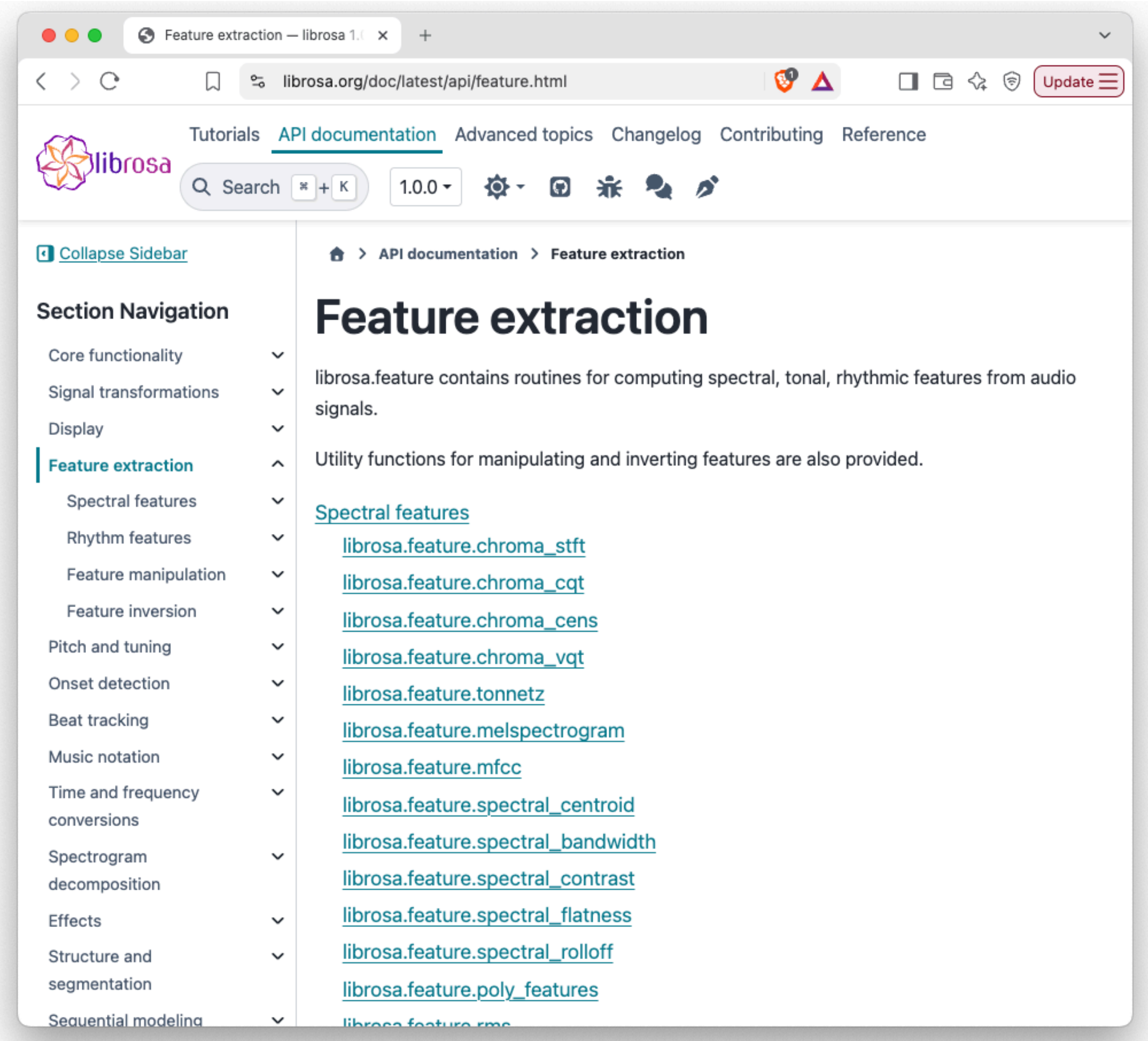
Root Mean Square
Zero Crossing Rate

Frequency Domain

Spectral Centroid
Spectral Bandwidth
Spectral Contrast
Spectral Flatness
Spectral Rolloff

Cepstral Domain

MFCC

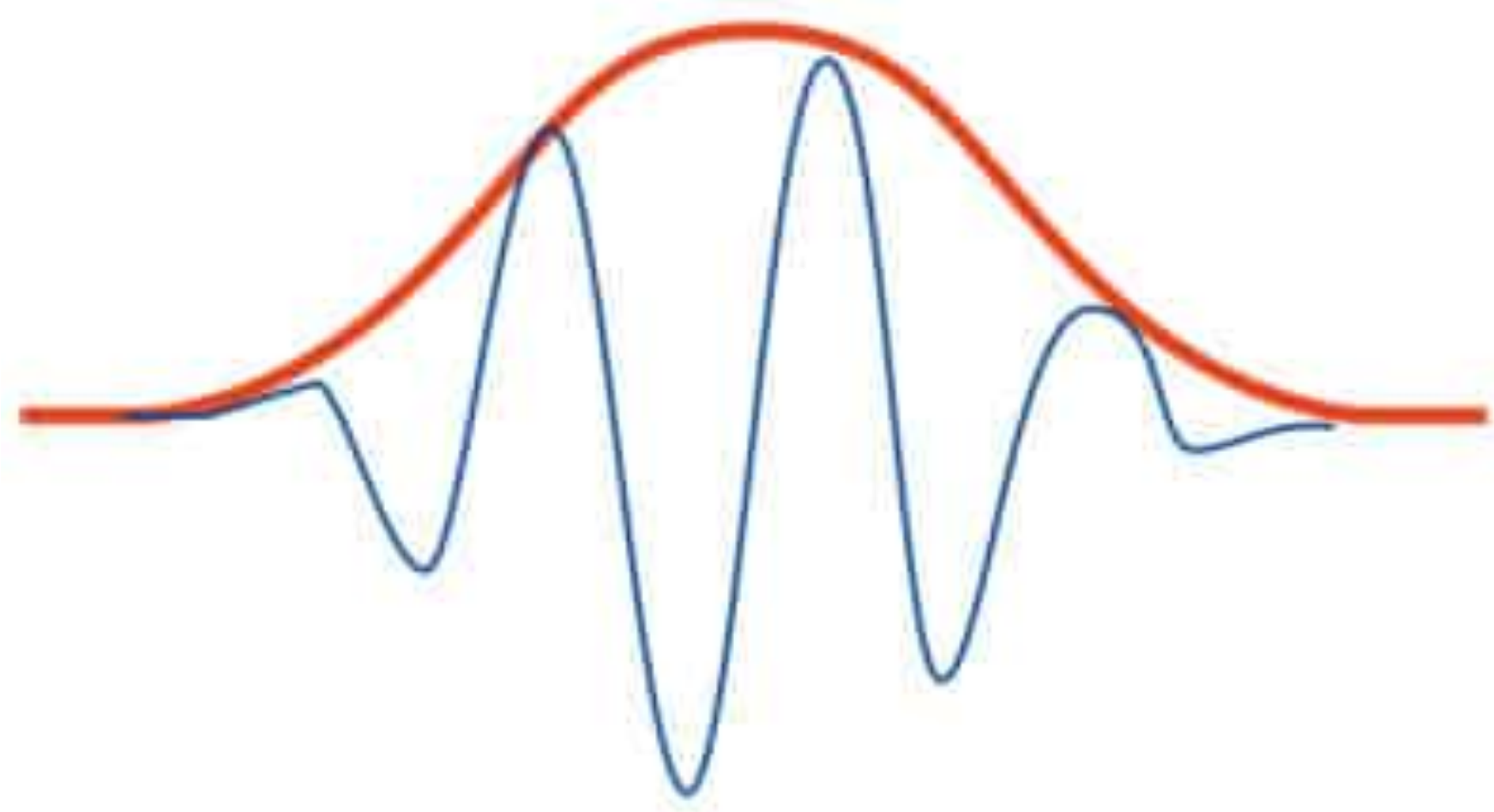


Librosa

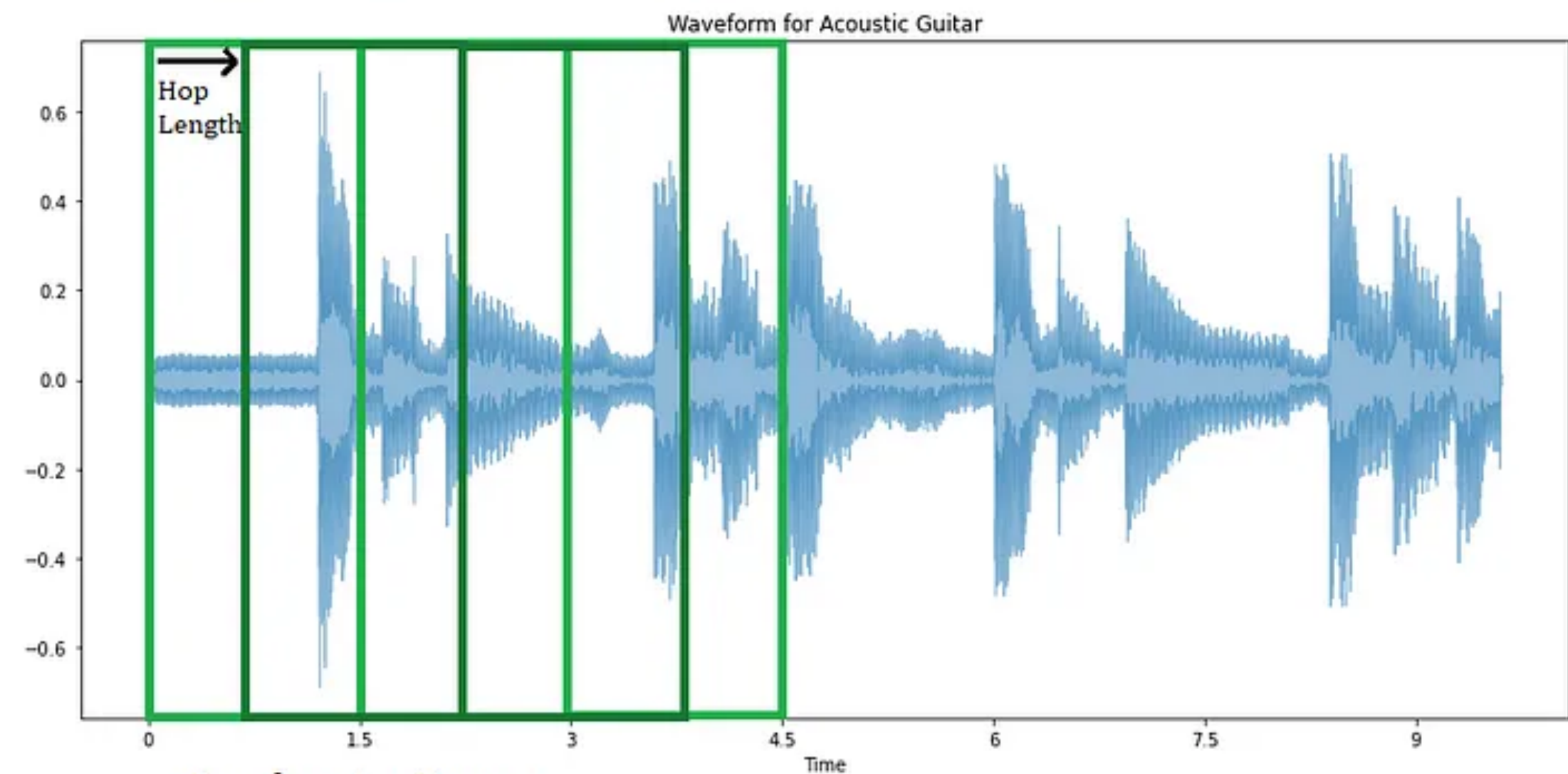
Audio Feature Descriptors - Basic Principle

$$RMS = \sqrt{\frac{\sum_{i=1}^n x_i^2}{N}}$$

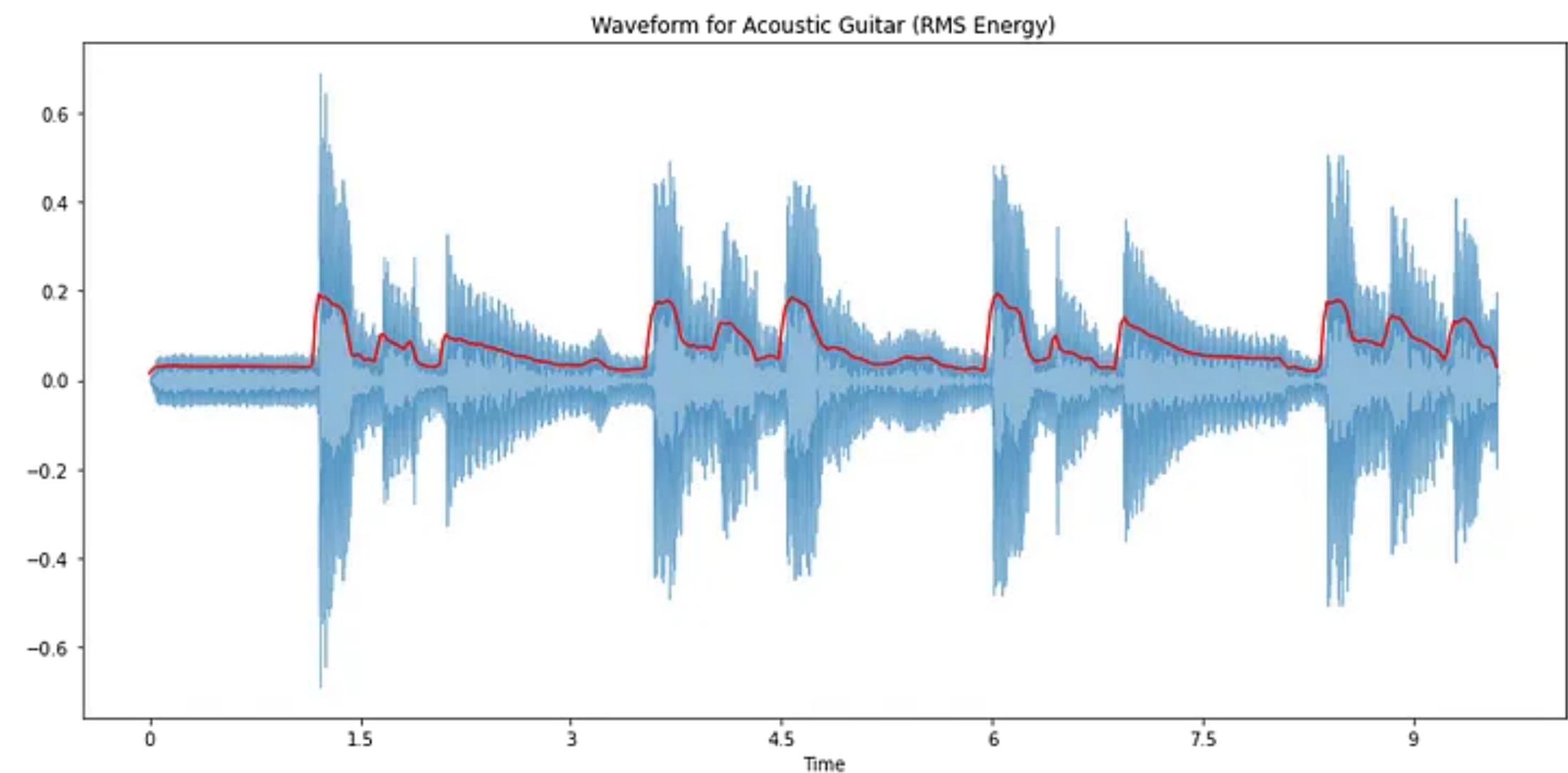
Descriptor Calculation



Audio Windowing



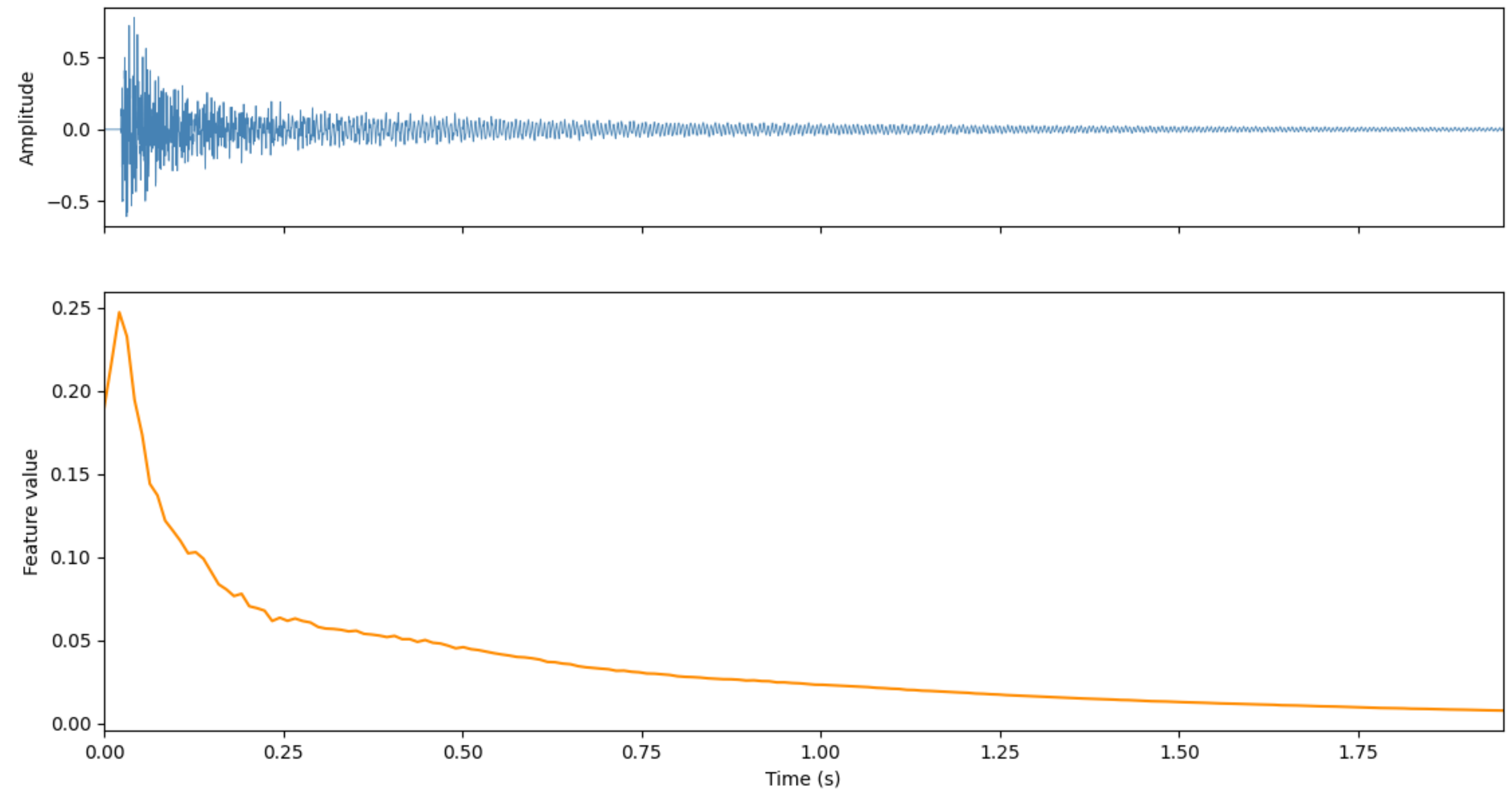
Overlapping Frames
[Hop Length = Frame Length/2]
Dark green denotes additional frames



Time Domain - Root Mean Square

RMS is used to measure the average loudness of a waveform

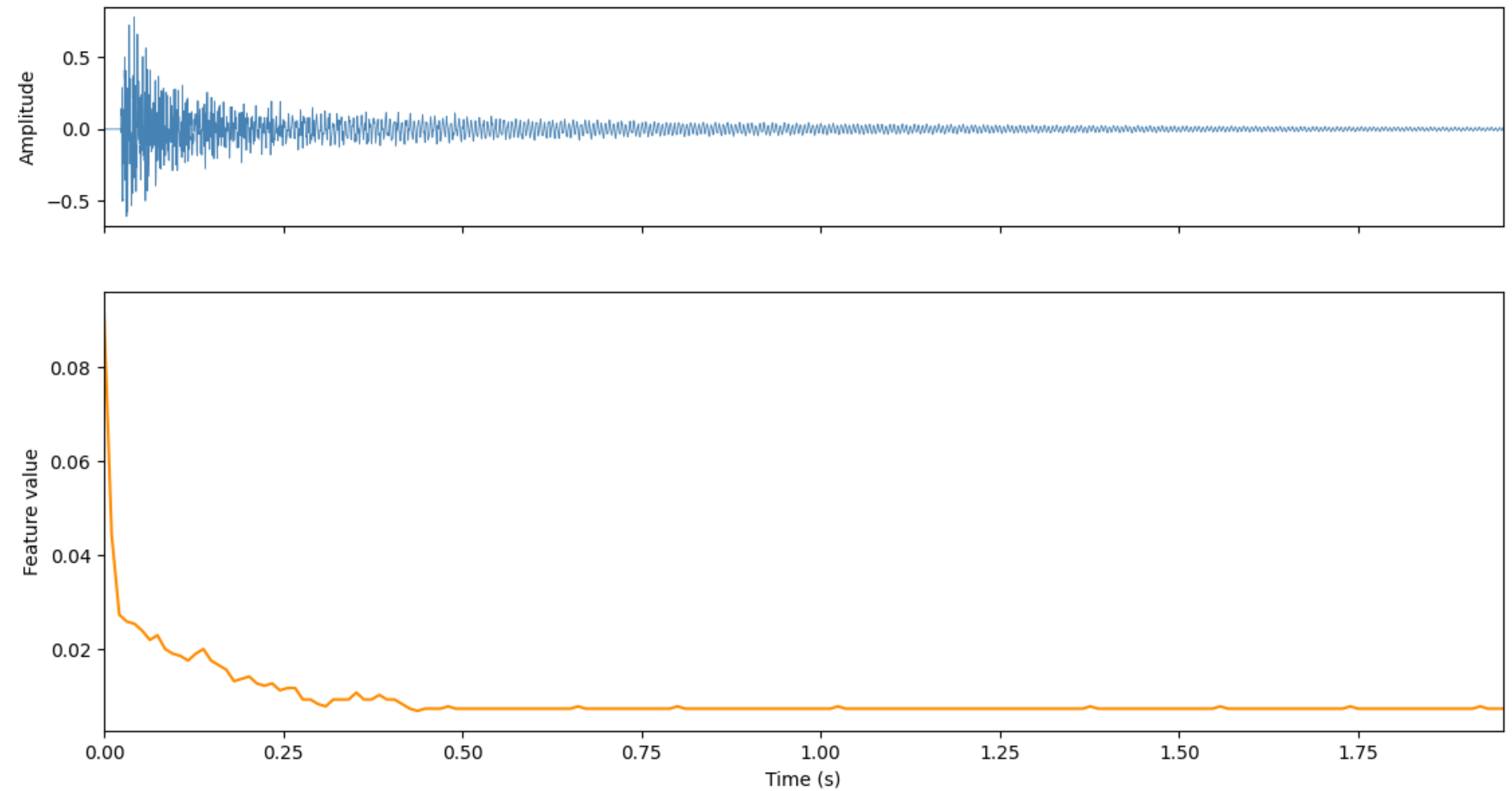
$$RMS = \sqrt{\frac{\sum_{i=1}^n x_i^2}{N}}$$



Time Domain - Zero Crossing Rate

Compute the zero-crossing rate of an audio time series.
Cheap proxy for noisiness / percussiveness.

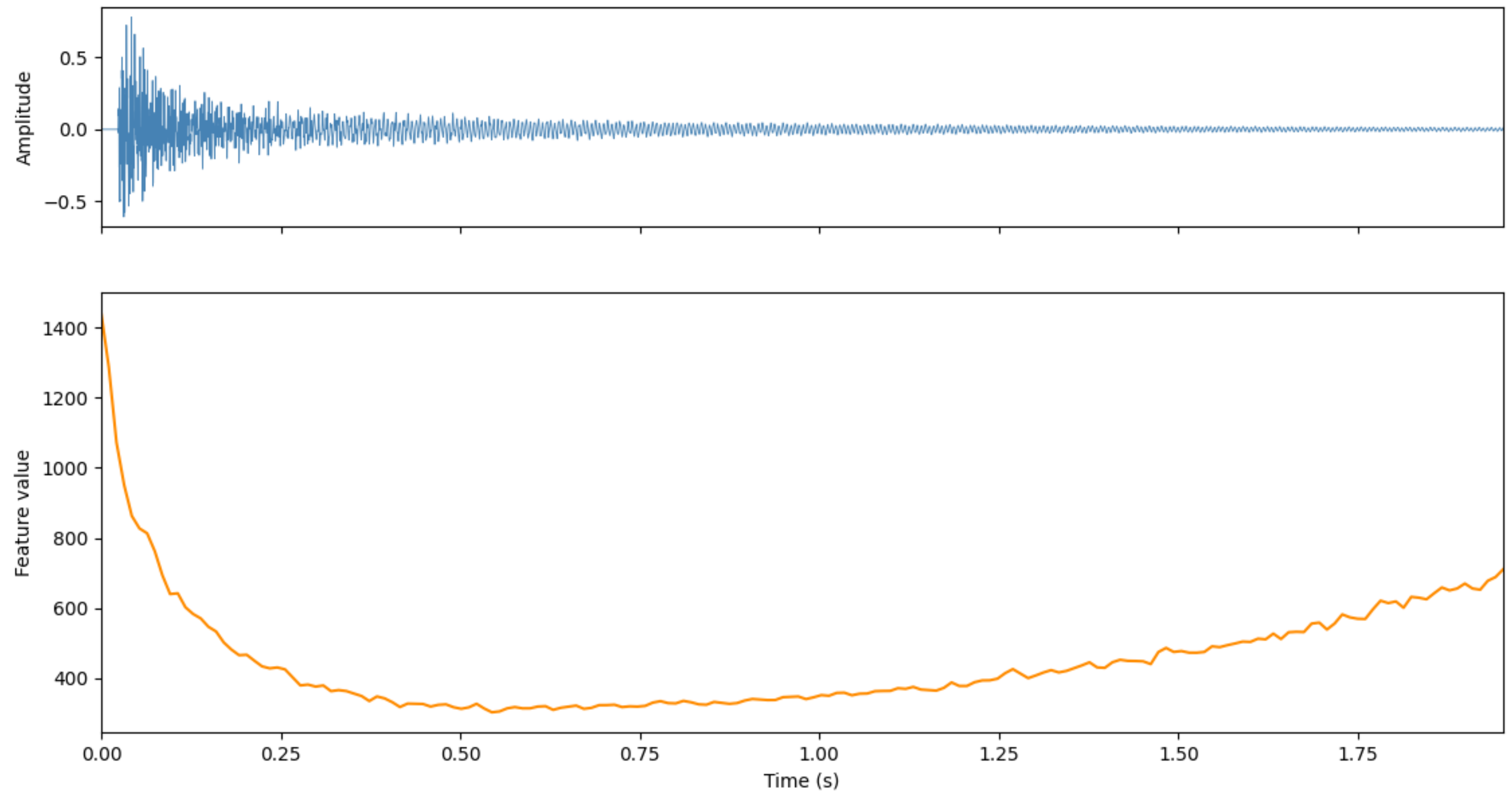
$$zcr = \frac{1}{T-1} \sum_{t=1}^{T-1} |\text{sgn}[s(t)] - \text{sgn}[s(t-1)]|$$



Frequency Domain - Spectral Centroid

The spectral centroid represents the "center of gravity" of the spectrum. It is used as an indication of brightness of a sound.

$$f_c = \frac{\sum_k S(k)f(k)}{\sum_k S(k)}$$

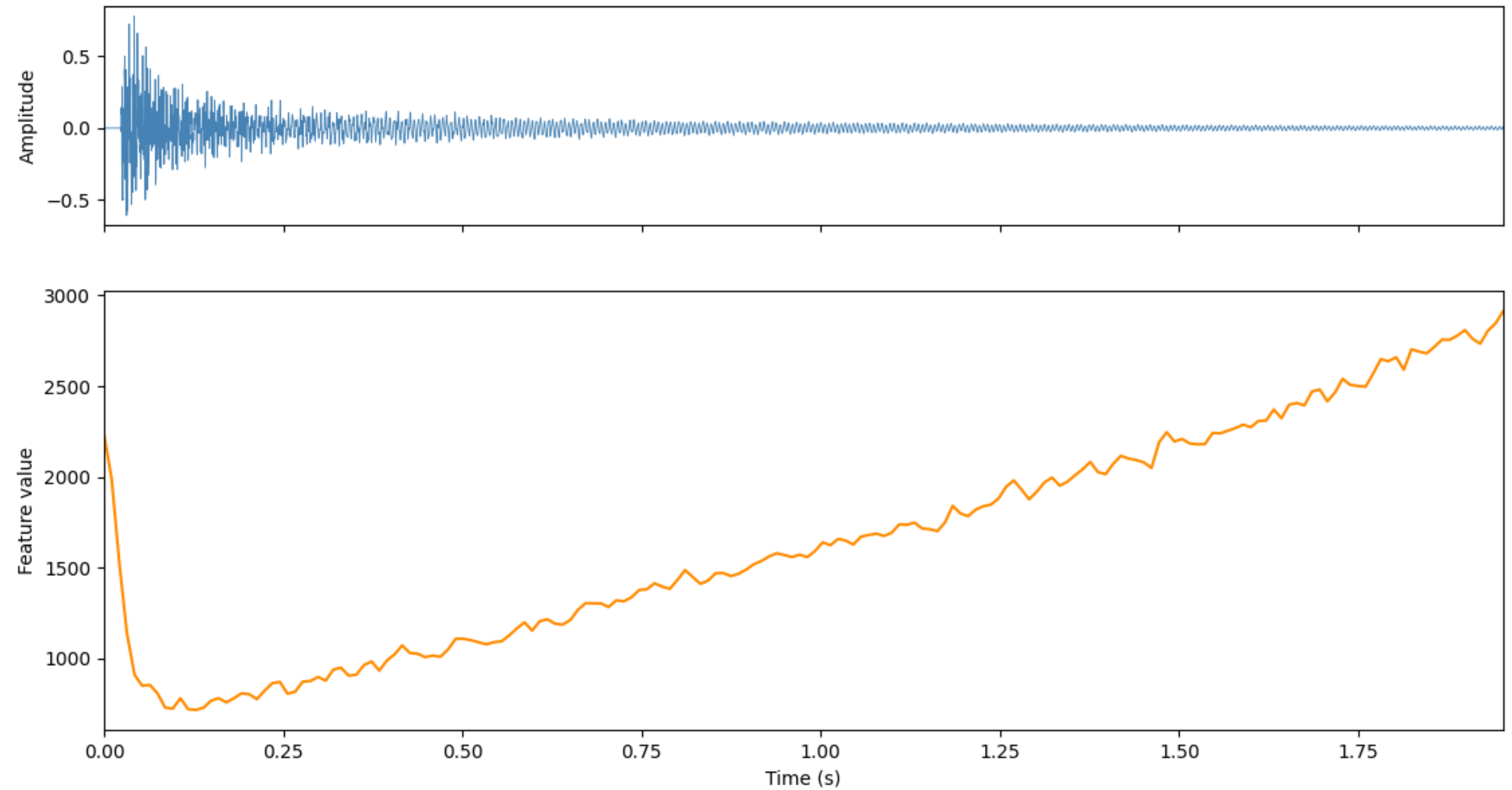


Frequency Domain - Spectral Bandwidth

Represents the spread of the spectrum.

It is used as an indication of the complexity versus purity of a tone.

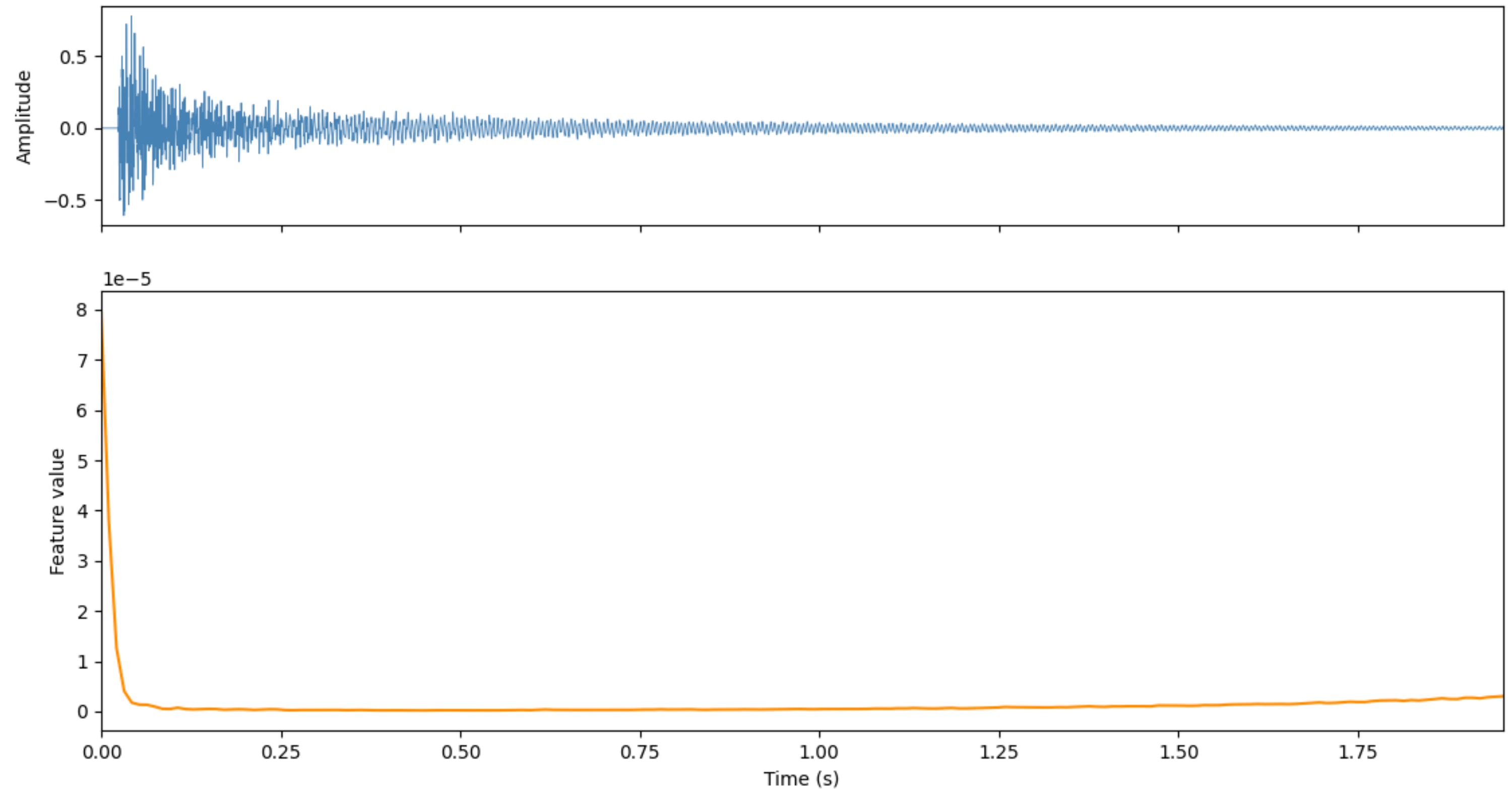
$$\left(\sum_k S(k)(f(k) - f_c)^p \right)^{\frac{1}{p}}$$



Frequency Domain - Spectral Flatness

Spectral flatness is an indication of the peakiness of the spectrum. A higher spectral flatness indicates noise, while a lower spectral flatness indicates tonality.

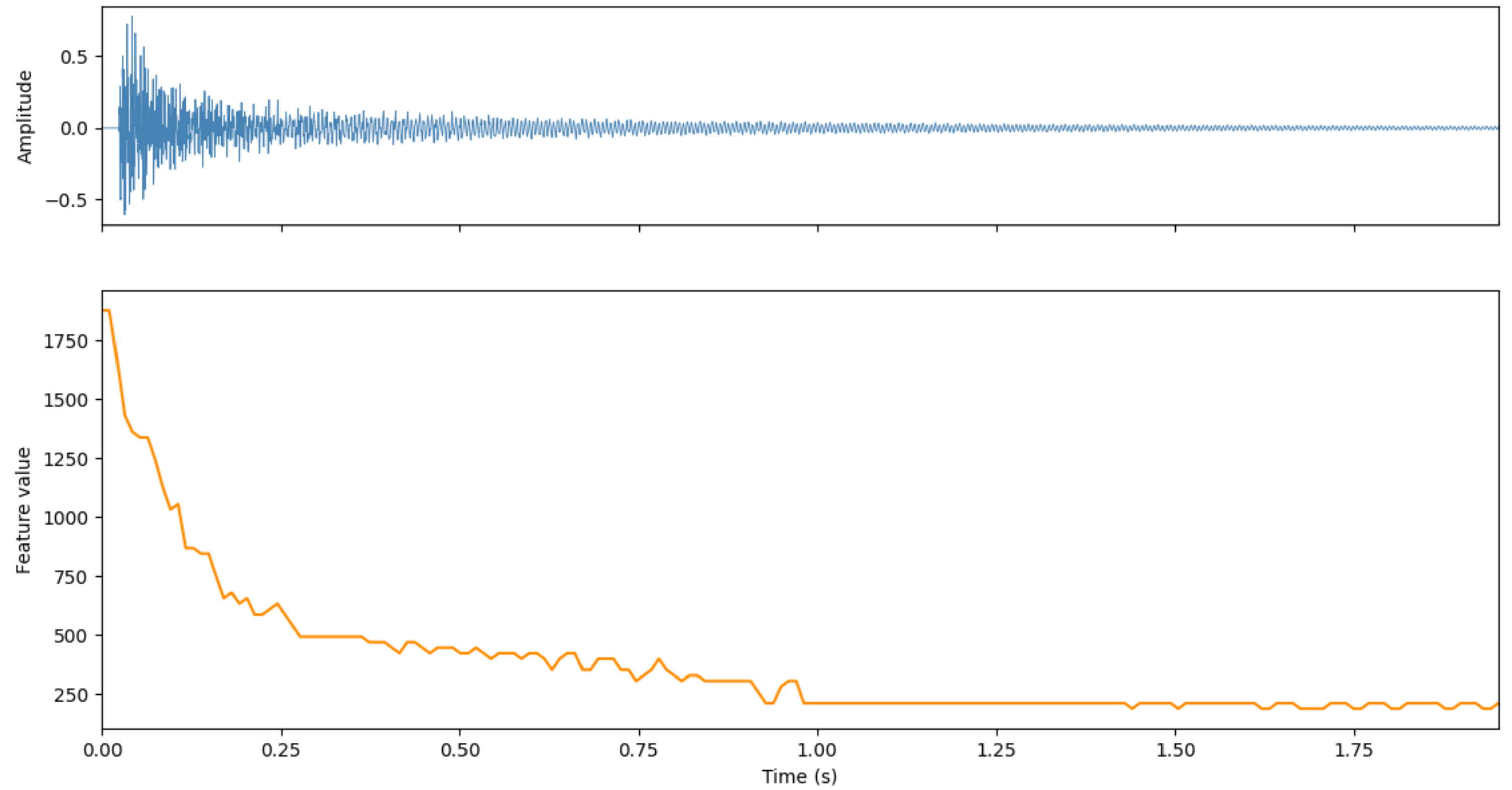
$$SF_k = \frac{\sqrt[\phi]{\prod_{i=1}^{\Phi} m_k(i)}}{\frac{1}{\Phi} \sum_{i=1}^{\Phi} m_k(i)} = \frac{\exp\left\{\frac{1}{\Phi} \sum_{i=1}^{\Phi} \log(m_k(i))\right\}}{\frac{1}{\Phi} \sum_{i=1}^{\Phi} m_k(i)}$$



Frequency Domain - Spectral Rolloff

Spectral rolloff is the frequency below which a specified percentage of the total spectral energy, e.g. 85%, lies. Low rolloff sounds have a “warm” or “muffled” timbre. High rolloff sounds possess a “bright” or “sharp” timbre.

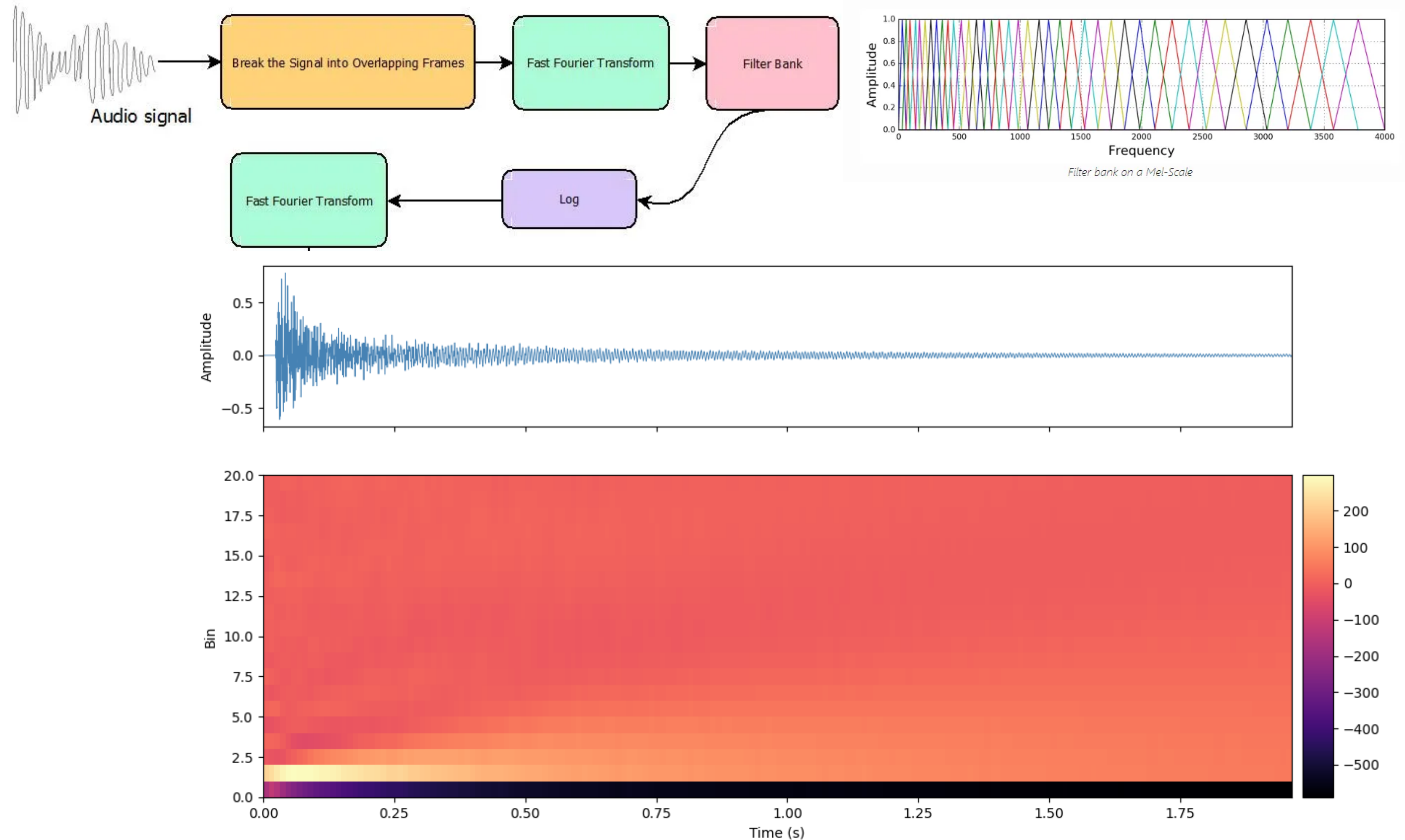
$$k_{\text{rolloff}}(t) = \min \left\{ k : \sum_{i=0}^k S_t[i] \geq \rho \sum_{i=0}^{K-1} S_t[i] \right\}$$



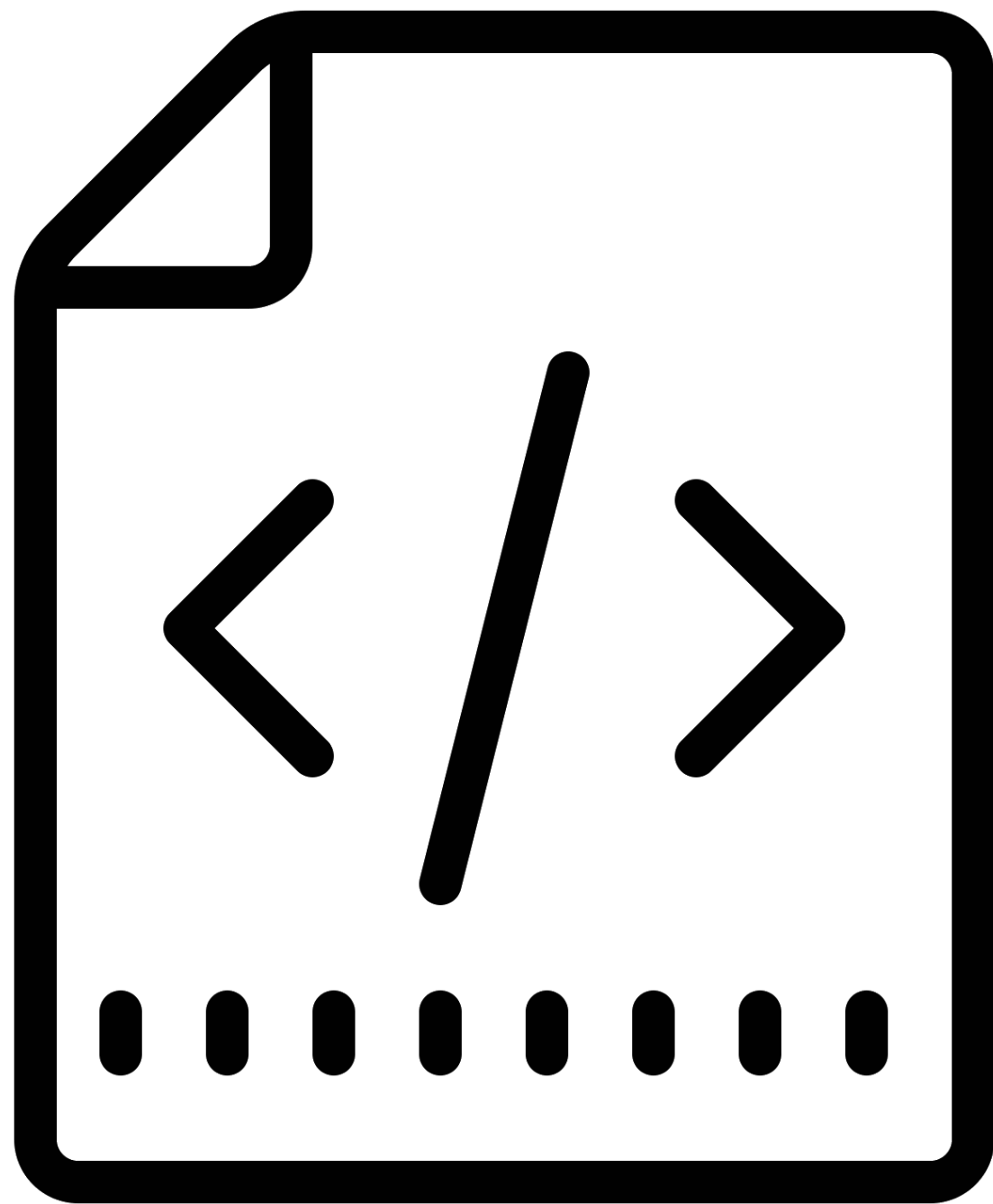
Cepstral Domain - MFCC

MFCCs are generated by taking the Fourier transform of a signal and then converting it to the mel frequency domain, then taking the logarithm of the power at each mel band and then taking the discrete cosine transform of the list of mel log powers, as if it were a signal. The MFCCs are the amplitudes of the resulting spectrum.

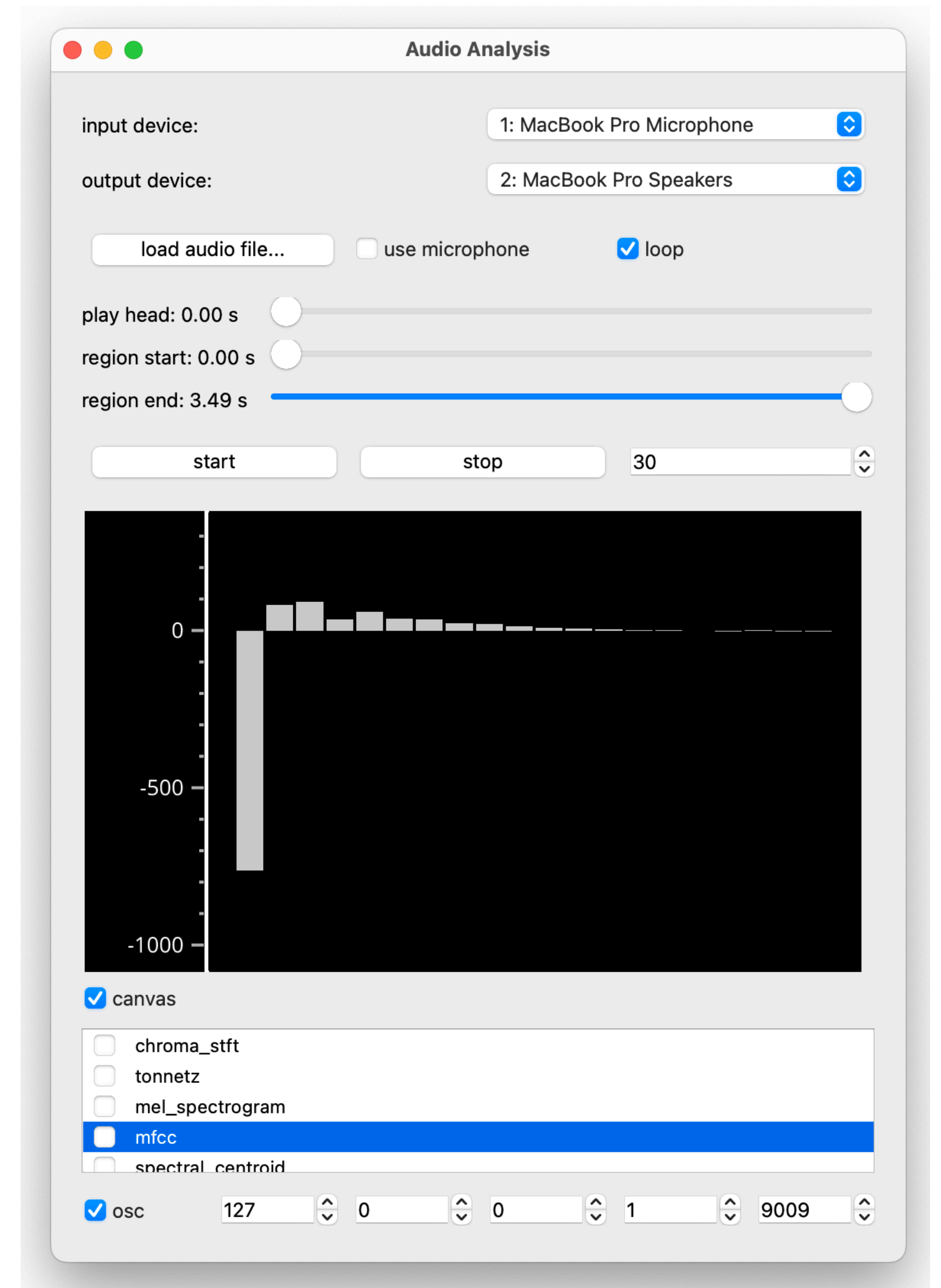
Compact representation of timbre.



Python Demo



AI-Sound / AudioAnalysis / audio_analysis_hello_world.py



AI-Sound / AudioAnalysis / audio_analysis.py